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കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

പൊതുവിദ്യാഭ്യാസ
വകുപ്പ്- കേരളം

ഡയറ്റ് കണ്ണൂർ

समग्र शिक्षा के रल
Samagra Shiksha Kerala
സമഗ്ര ശിക്ഷാ കേരളം



എസ് എസ് എൽ സി
വിദ്യാർത്ഥികൾക്കുള്ള
പഠനപിന്തുണാ സഹായി

2

ബയോളജി

ENGLISH MEDIUM

Smile 2026

Special Module to Improve achievement Level in public Exams

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സ്മൈൽ റിസോഴ്സ് ഗ്രൂപ്പ് അംഗങ്ങൾ

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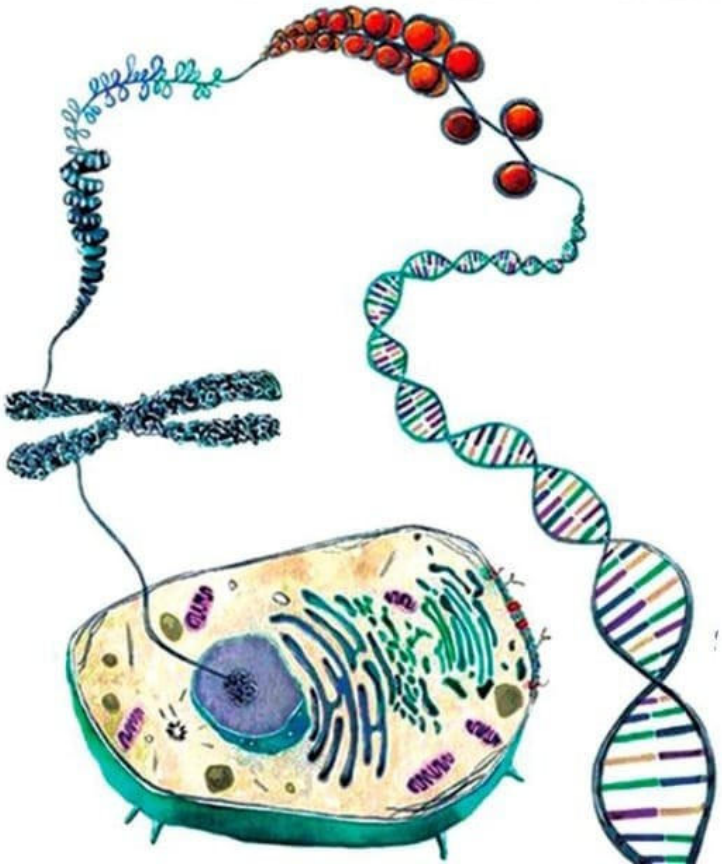
ജില്ലാവിദ്യാഭ്യാസപരിശീലനകേന്ദ്രം ഡയറ്റ്, കണ്ണൂർ

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SMILE 2025-26

BIOLOGY



ENGLISH MEDIUM

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PREFACE

During the last academic year, students from the Kannur district achieved excellent success at the state level in the 10th standard examinations. It is the duty of each one of us to maintain and further improve this success in the coming years as well. Recognizing its importance, the Kannur District Panchayat has decided to implement the 'Smile' project this year in a timely manner. We have attempted to prepare a material incorporating activities that will enable students to achieve high grades in Biology, which holds significant importance among science subjects. To achieve the learning objectives in Biology, students need to go through learning experiences and activities. For this, concepts from the textbook have been explained and model questions have been prepared. The aim of this initiative is to guide students to a high standard, while working within the time allotted for science subjects.

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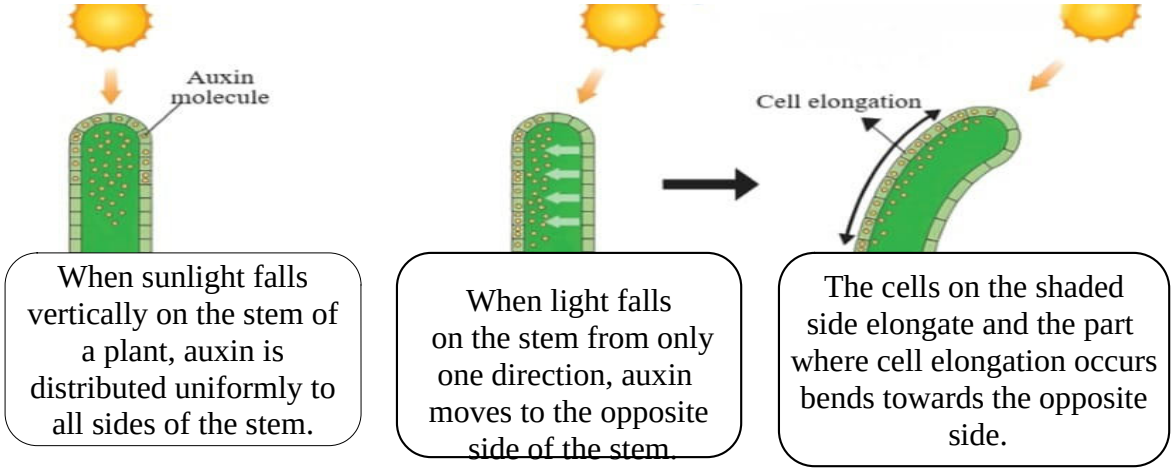
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Unit - 4
Chemoreception in Organisms

Hormones and Life processes

- A cell in which a hormone acts is called the target cell.
- The target cell of each hormone has receptors suitable for that hormone.
- Receptors are seen in the cell membrane and within the cells.
- Hormones combine with their receptors to form a hormone-receptor complex.

Action of Auxin



Auxin was discovered by F W Went

Plant Hormones and their Functions

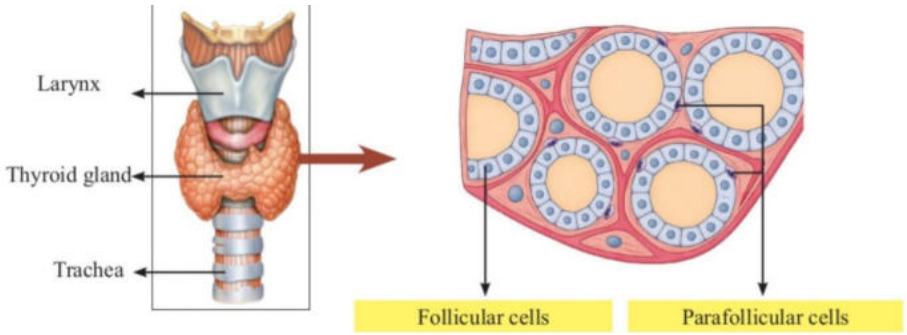
Hormone	Functions
Auxins	• Auxins increase the length of the stem and inhibit lateral bud growth . • regulate tropic movements. • stimulate the formation of flowers and fruits.
Gibberellins	• influence the growth of fruits and seeds. • break seed dormancy. • stimulate cell division in both stems and roots.
Cytokinins	• stimulate cell division and cell differentiation in germinating seeds. • Stimulate cell division in both stems and roots. • stimulate the growth of lateral buds . • promote the transport of nutrients to growing regions.
Ethylene	• ageing of leaves and flowers • ripening of fruits.
Abscisic acid (ABA)	• inhibits the growth of plant parts • to tide over unfavourable conditions. • Promotes abscission of mature leaves and fruits.

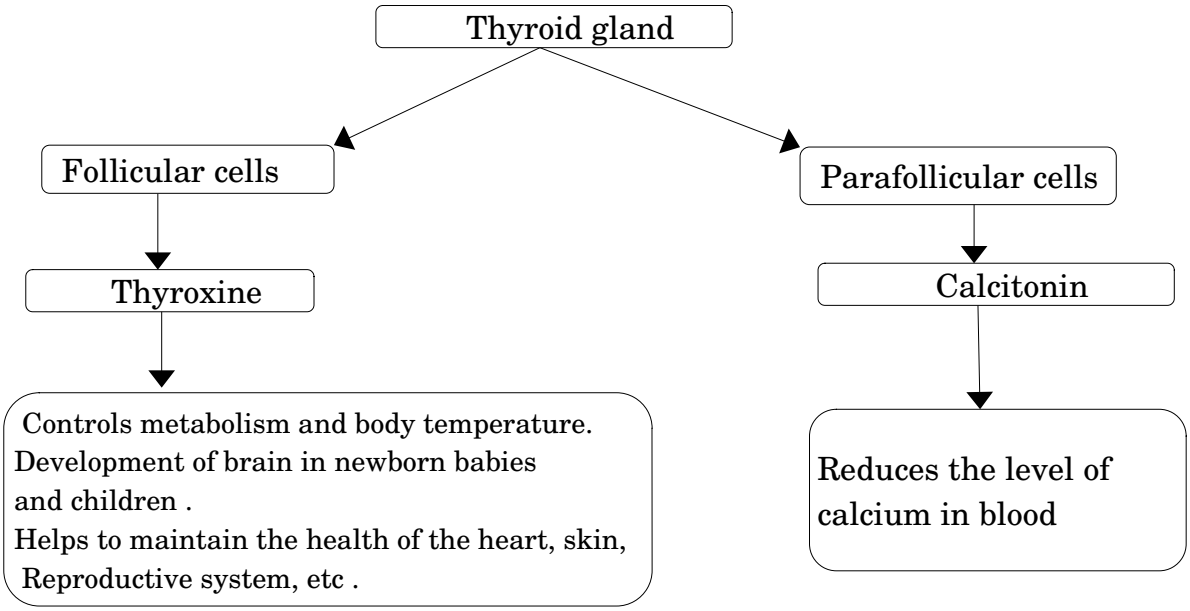
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Synthetic Plant Hormones and their Uses

Synthetic Plant hormone	**Uses**
Synthetic auxins	• Used to induce root formation in stem cuttings. • Also used as a weedicide (2,4-D)
Synthetic gibberellins	• To increase the size of fruits in grapevines . • to promote stem elongation in sugarcane.
Synthetic ethylene	• used to make fruits ripen uniformly for harvest at the same time. • used to ripen bananas and tomatoes simultaneously.

Thyroid gland





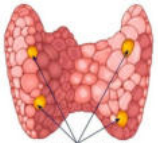
Defects related to thyroid gland

Condition	Causes	Symptoms
Hyperthyroidism	Excessive secretion of thyroxine.	• Increase in the rate of metabolism . • weight loss, heart beat increases, excessive sweating, anxiety .

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Hypothyroidism	Reduced secretion of thyroxine	• Metabolism decreases. • Body weight increases . • heart rate slows down, fatigue, intolerance to cold.
Goitre	Various disorders of thyroid gland deficiency of iodine.	• Swelling in the neck region. • difficulty in swallowing

Parathyroid gland



Parathyroid gland

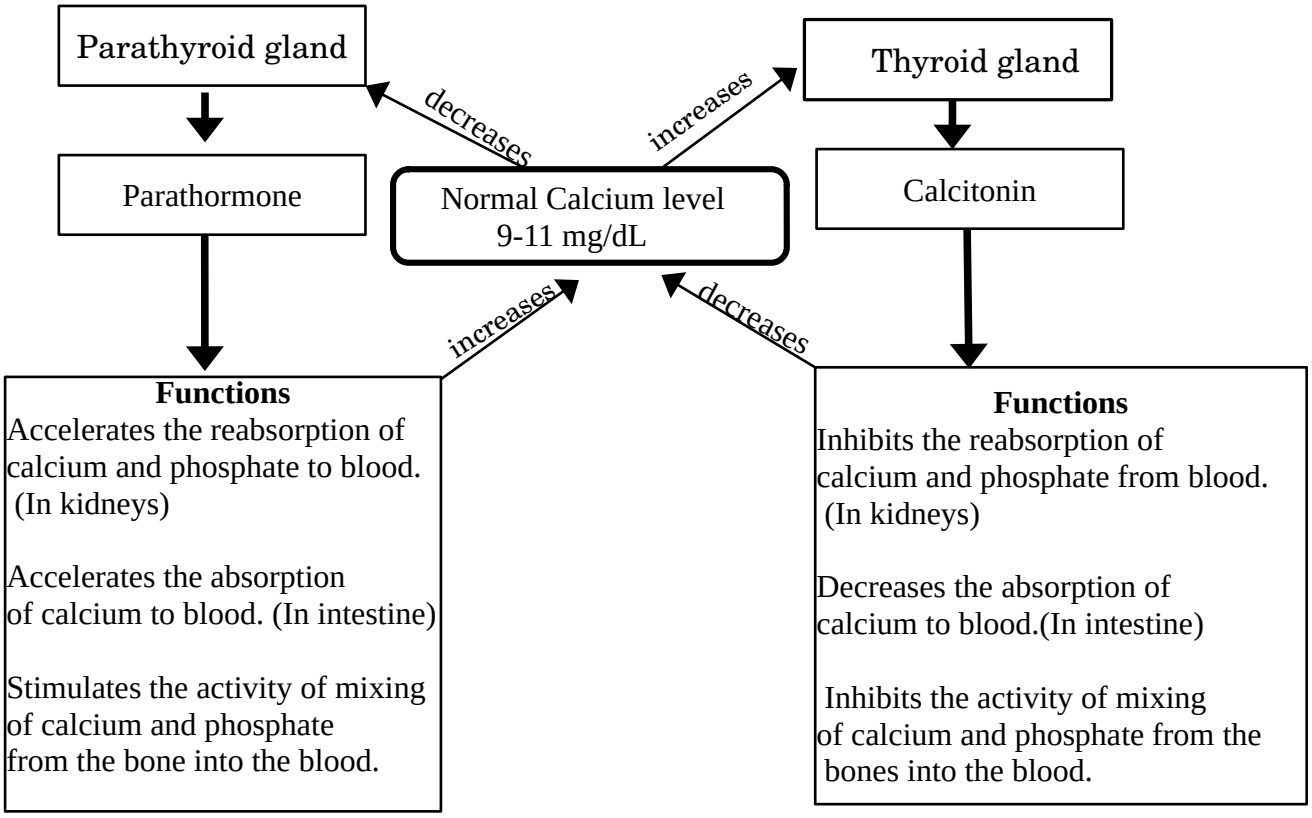
Parathyroid gland located behind the thyroid gland

parathormone produced by **pathyroid gland.**

Regulation of calcium in blood

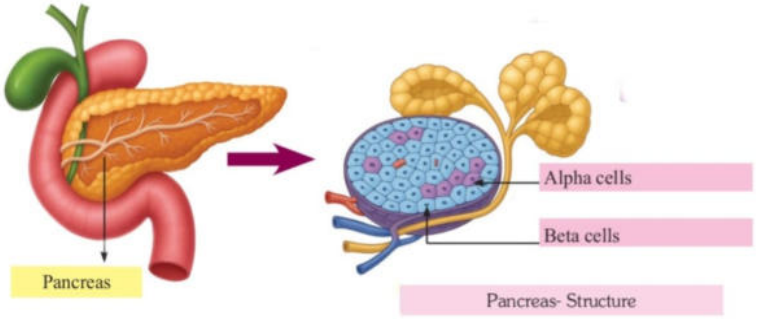
When the level of calcium decreases in blood

When the level of calcium increases in blood

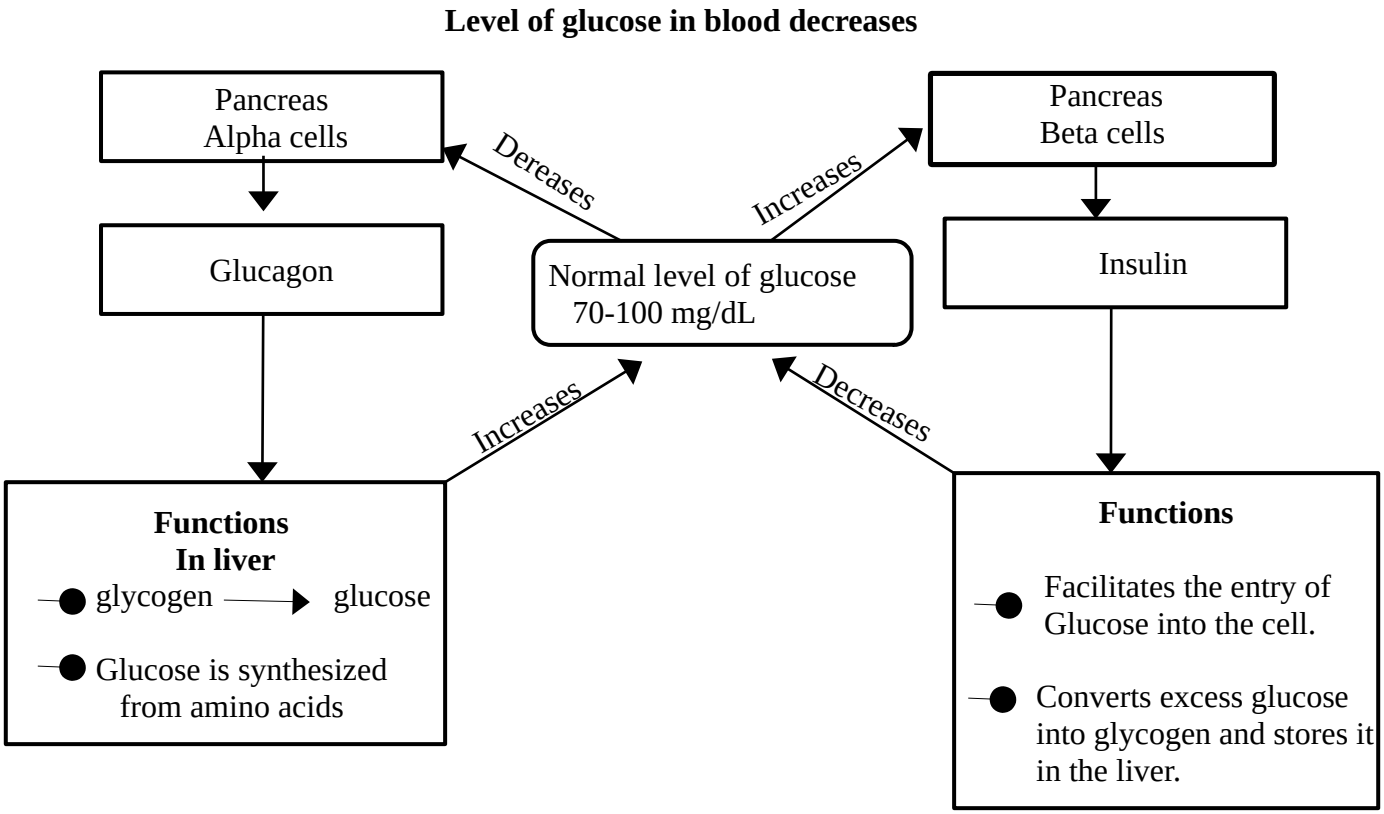


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Pancreas



Regulation of glucose in blood



Diabetes

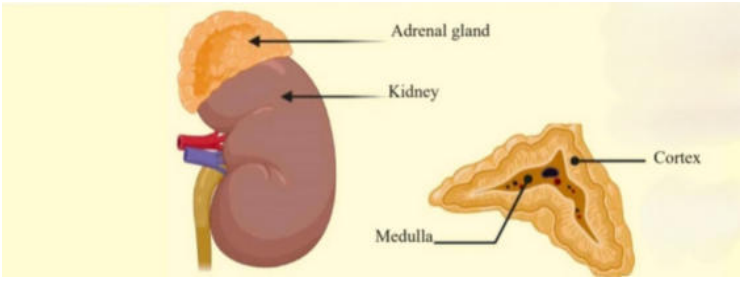
- Diabetes mellitus is a condition in which the level of glucose in blood rises above normal.
- FBS above 126mg/dL
- Excessive thirst, hunger, frequent urination, weight loss and fatigue are the symptoms.
- If diabetes is ignored, it can lead to heart disease, stroke, renal diseases and vision problems.
- Early identification of diabetes and preventive measures to avoid the diseaseas well as making life style changes is crucial in healthcare.

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Disease	Causes
Type – 1 Diabetes	Insulin production capacity is lost due to the destruction of beta cells.
Type – 2 Diabetes	Variation in insulin production, lack of dietary control, and decreased effectiveness of insulin.

Adrenal gland

- Overcoming emergency situations
- Some hormones produced by the adrenal gland which is part of the endocrine system also functions along with the sympathetic system.



➢ **Outer part – Cortex**

➔ The hormones produced by the cortex are called cortico steroids (cortisol, aldosterone, and androgens).

➢ **Inner part - Medulla**

➔ The hormones produced by the medulla are epinephrine and norepinephrine.
➔ These hormones prepare the body to deal with an emergency situation through a number of physiological changes such as an increased heart rate and blood pressure, increased blood glucose levels, increased blood flow to the heart and muscles, decreased blood flow to the skin and digestive organs, etc.

Corticosteroids and functions

Hormones	Function
Cortisol (glucocorticoids)	• Maintains glucose homeostasis • Suppresses the Immune responses • Resists inflammation and allergy of the body
Aldosterone (mineralocorticoids).	• Helps in maintaining equilibrium of minerals in the body • Regulates blood pressure
Androgens (gonadocorticoids)	• Controls sexual development and sexual behaviour.

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Pineal gland

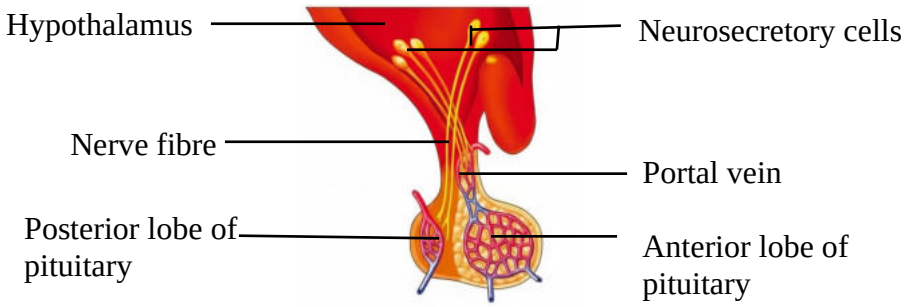
Sleep and Awakening

- The pineal gland, located near the thalamus in the human brain,
- It produces a hormone called **melatonin**.
- The variations in the secretion of this hormone influences sleep and wakefulness.
- Melatonin controls the activities that occur at regular intervals through out the day therefore, pineal gland is known as the biological clock.
- Using a mobile phone for a long time at night reduces melatonin production and disturbs sleep.

Thymus gland

- Located behind the sternum.
- The hormone **thymosin** produced by **thymus** gland
- Plays a major role in the body's immune system.
- helps in the maturation of **T lymphocytes.**

Hypothalamus and Pituitary



Hormones secreted by the neurosecretory cells of the hypothalamus.

1.Hormones that reach the anterior lobe of the pituitary through the portal vein:

- Releasing hormones - stimulate the production of tropic hormones.
- Inhibitory hormones - Inhibit the production of tropic hormones.

2. Hormones that reach the posterior lobe of pituitary through nerve fibres:

- Oxytocin – lactation (production of milk) and contraction of smooth muscles in the uterus.
- Vasopressin – helps to maintain water balance in the body.

Hormones produced by the anterior lobe of pituitary

1.TSH - Stimulates Thyroid gland.
2.ACTH- Stimulates Adrenal cortex.
3.GTH – Stimulates the action of ovary and testes.
4. Somatotropin – Helps in body growth.
5.Prolactin – Produces milk.

Growth hormone(Somatotropin,GH)

It produced by the anterior lobe of the pituitary gland .

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Growth disorders

Disorder	Causes	Symptoms
Dwarfism	The production of somatotropin decreases during the growth phase.	Stunted growth.
Gigantism	The production of somatotropin increases during the growth phase.	Excessive body growth
Acromegaly	The production of somaotropin increases after growth phase	Excessive growth of body parts such as palms, foot and jaws.

vasopressin (Antidiuretic hormone – ADH)

During summer

When the production of vasopressin increases

↓

the reabsorption of water from the kidneys increases

↓

the amount of urine decreases.

During Rainy Season

When the production of vasopressin decreases

↓

the reabsorption of water from the kidneys decreases

↓

the amount of urine increases.

Reproductive Hormones

Hormone	Gland	Function
Testosterone	Testes	• The activity of male sex organs • Sperm production • Controls secondary sexual characteristics
Estrogen	Ovary	• The activity of female sex organs. • Production of ovum • Menstrual cycle • Control of secondary sexual characteristics

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Progesterone	Ovary	• Maintaining the foetus in the uterus • Menstrual cycle
Prolactin	Pituitary (anterior lobe)	• Produces milk
Oxytocin	Pituitary (posterior lobe)	• Lactation (Production of milk) • Contraction of the smooth muscles in the uterus.

Mental health

- The chemical messengers in the nervous system, known as neurotransmitters.
- Neurotransmitters such as dopamine, serotonin and endorphins also function as hormones.

Pheromones

Pheromones are chemical substances that are secreted by organisms to their immediate surroundings to facilitate the communication with the same species .

Different type of pheromones	Function
Territorial pheromone	To determine the territory
Aggregation pheromone	To ensure the availability of food
Alarm pheromone	Signalling dangers
Trail pheromone	For trail marking
Sex pheromone	To attract mates

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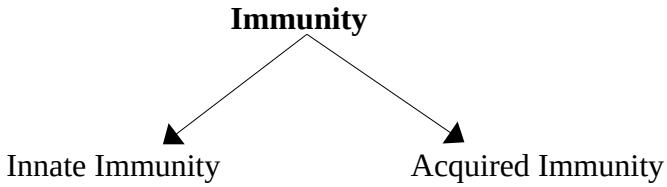
Unit 5

Immunity and Healthcare

Disease

Any change in the homeostasis of the body is called disease.

Antigens: Anything, including pathogens which enter the body and cause diseases are called antigens.
Eg: pathogens, Pollen grains, dust, certain food items



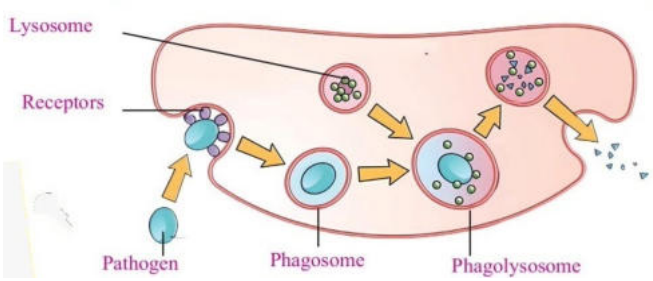
- The natural immune mechanism that exists in the body by birth is called **innate immunity.**
- Innate immunity includes various mechanisms to prevent antigens from entering the body, these are generally called **primary level immunity.**
- The pathogens bypass the primary level immunity and enter the body, the innate immune system that defends against them is called the secondary level immunity .

White blood cells and their functions

White blood cells	Functions
 Neutrophil	• It engulfs and destroys bacteria • produces chemicals against pathogens .
 Eosinophil	• Neutralizes chemicals released when there is an allergy. • destroys parasites.
 Monocyte	• It engulfs and destroys foreign substances, bacteria, and dead cells. • stimulates other Immune cells.
 Basophil	• It produces heparin, which prevents blood clotting. • histamine which causes inflammation in cases of allergy.

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1. **Phagocytosis**



- Phagocytosis is the process by which white blood cells engulf and destroy pathogens.
- The blood cells that take part in phagocytosis are called Phagocytes.
- Neutrophils and Monocytes are phagocytes

2. **Inflammatory response**

- It is the process by which blood flow increases to the injured or bruised area to bring white blood cells to fight an infection.
- The cells at the injured part release histamine and other chemical substances. .
- This dilates the blood vessels .
- As blood flow increases, monocytes and neutrophils reach the injured area. They engulf and destroy the pathogens.

3. **Fever**

- Fever is the condition that occurs when the body temperature rises above the normal range 98.6^{o}F or 37^{0}C .
- When infection or inflammation occurs, white blood cells release chemical substances called pyrogens.
- These trigger the brain to raise the body temperature.
- High temperature helps phagocytes to destroy pathogens more quickly by engulfing them, inhibit the growth of pathogens, and enhance immunity..

4. **Clotting of blood**

a) Platelets ⟶ Thromboplastin

b) Thromboplastin + Prothrombin ⟶ Thrombin

c) Thrombin + Fibrinogen ⟶ Fibrin

d) Fibrin + Blood cells ⟶ Blood clot

5. **Wound healing**

- As blood clots and inflammation occur at the wound, new tissues are formed leading to wound healing.
- If the wound is healed with connective tissue replacing the lost tissues, a scar will remain.

Non-specific Immunity : All innate immune responses are irrespective of the type of pathogen or their actions.

Acquired Immunity/ Specific Immunity

- This is the immunity that develops after birth.

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Specific immunity : This immune mechanism specifically recognises and defends antigens (like the pathogens) that enter the body.

- This immune defence is primarily carried out by lymphocytes**.**
- There are two types of Lymphocytes -T lymphocytes , B lymphocytes

Cells in specific Immunity	**B lymphocytes**	**T lymphocytes**
Formation	Bone marrow	Bone marrow
Maturation	Bone marrow	Thymus gland
Function	Produce proteins called antibodies which act against antigens.	Identify and destroy the infected cells and cancer cells.

Artificial Immunity

- Immunisation is an artificial method of preparing the immune cells in advance, expecting the attack of pathogens.
- The immunity acquired through immunization is called **artificial immunity**

Vaccines and diseases

Vaccine	**Components of vaccine**	**Preventable diseases**
MMR	Contains neutralised pathogens.	Measles, Mumps, Rubella (German measles)
OPV	Contains dead pathogens	Polio
HPV Vaccine	Contains parts of pathogens	Cervical cancer
COVID-19 Vaccine	Contains parts of the genetic material of the pathogen	Covid 19

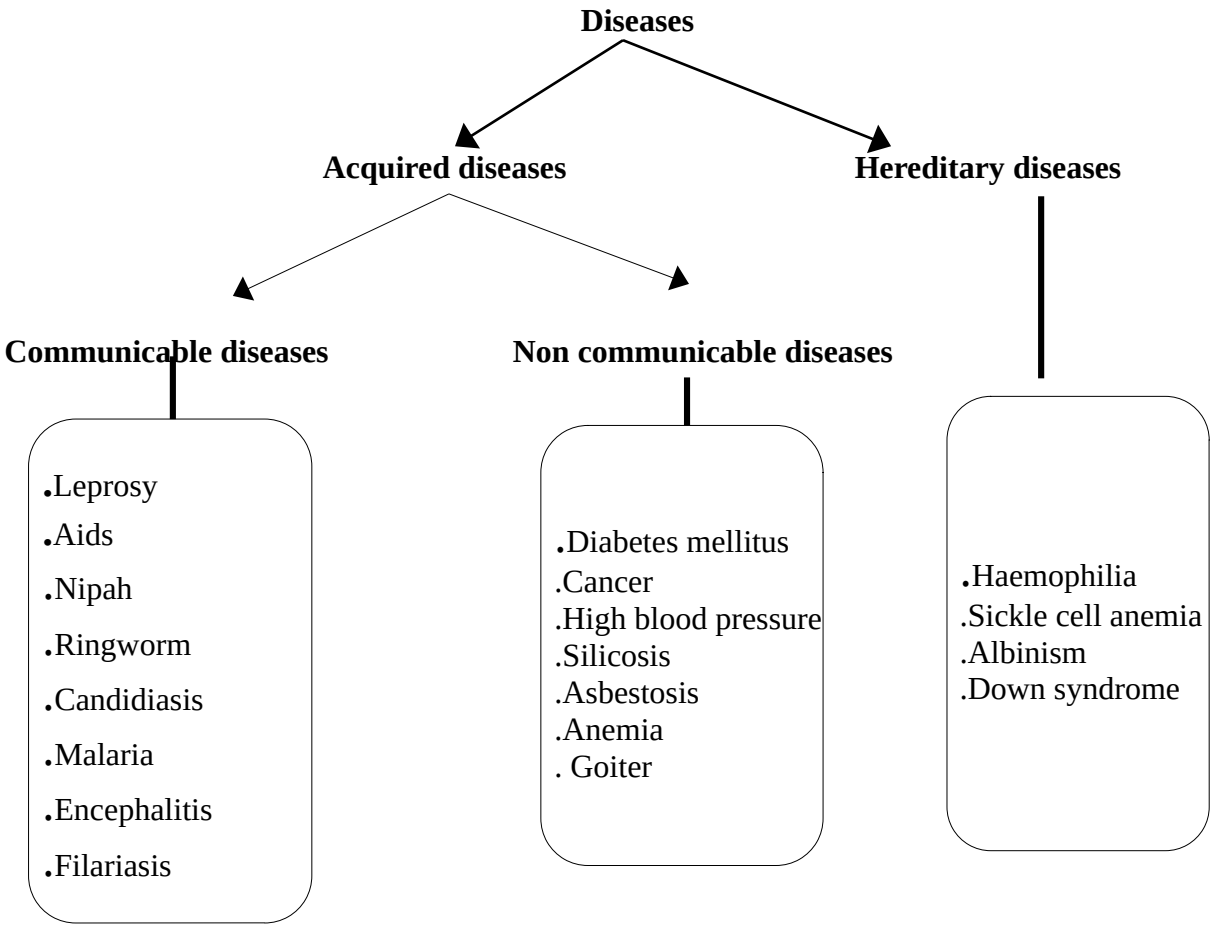
Mechanism of the Action of Vaccines

- Vaccines act as antigens in our body.
- It stimulates our immune system and helps to produce antibodies against the pathogens.
- These antibodies remain in the body and provide long-term protection against diseases.
- By taking the doses accurately vaccines help to gain protection from a disease for many years or even for a lifetime.

Diseases

- Malfunction of the immune system, genetic defects, unhealthy lifestyle, abundance of pathogens etc. Disrupt the homeostasis of the body and cause diseases.

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Modes of disease transmission

- Through surfaces
- Through direct contact
- Through contaminated water and food.
- Through droplets
- Through disease-carrying agents.

Disease	Pathogen
Bacteria	Tuberculosis, Rat fever.
Virus	AIDS , Nipah.
Fungus	Ring worm, Candidiasis.
Protozoa	Malaria, Amoebic encephalitis.
Worms	Filariasis,Worm infestation .

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Diseases	Pathogen	Mode of transmission	Part of the body affected	Symptoms	Prevention and treatment
Tuberculosis	Mycobacterium tuberculosis	when the patient is coughing, sneezing or talking	The lungs, kidneys,bones, and the brain	persistent cough , fever, loss of appetite, weight loss, fatigue and blood in the sputum.	BCG vaccination. Long-term treatment by using multiple antibiotics (DOTS) simultaneously
Rat fever (Leptospirosis)	Leptospira	Through wounds Through the mucus membranes of the eyes and nose .	Kidneys, Liver, Lungs	Sudden onset of fever, headache, muscle pain, redness of the eye, vomiting	Use preventive medicines Treatment using antibiotics
AIDS	Human Immunodeficiency Virus (HIV)	HIV can be transmitted through sexual contact with an infected person. By sharing needles and syringes that are not HIV-free. Through transfusion of blood or organs containing HIV. From an infected mother to the foetus.	T lymphocytes	Pneumonia, Tuberculosis, some types of cancer.	Antiretroviral Therapy (ART)
Nipah	Nipah virus	Through the saliva and urine of bats, through the remains of		Fever, head ache, muscular pain, cough, and	Only supportive care is available to reduce the

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		fruits eaten by bats, and from them to pigs.		vomiting are the initial symptoms As this becomes severe, disorientation, epilepsy, and loss of consciousness	symptoms and complications of the disease. Currently no antiviral medicines or vaccines
Ringworm		Through the infected persons,contact, contaminated objects and from animals		Circular reddish rashes, itching	
Candidiasis		Direct contact, through sharing of clothes, through sexual contact		White patches in the mouth, itching	
Malaria	Plasmodium	Through female Anopheles mosquitoes.		Intermittent fever, chills and shivering, head ache,vomiting, muscular pain, body pain and fatigue.	Anti malarial medicines
Amoebic encephalitis	Naegleria fowleri	Infection occurs when contaminated water enters the nose and the pathogens reach the brain through the nose.			
Filariasis	Filarial worms – mosquito	Culex mosquito	Lymphatic system	The legs, arms, and genital organs affected by the disease excessively swollen and become enlarged.	Avoid mosquito bites
Worm		By drinking		fatigue, lack of	Albendazole

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infestation		contaminated water, eating without washing hands, or biting nails.		enthusiasm, tiredness, lack of concentration, nutritional deficiency, loss of appetite, weight loss, stomach pain, dizziness, and vomiting.	tablets

Non contagious disease.

Cancer

- Cancer is a condition in which abnormal and uncontrolled cell division leads to the multiplication of cells. This spreads to other parts of the body.
- When the control mechanisms of the cell division process fail, normal cells turn into cancer cells.
- Cancer cells spread to other parts of the body through blood and lymph.

Causes of cancer

genetic changes , environmental factors , viral infections ,life style

Treatment method

- Surgery, Chemotherapy, Radiation therapy, Immunotherapy, Targeted therapy , Hormone therapy.

Non contagious diseases

Type	Diseases	Reason
Lifestyle diseases	• Diabetes mellitus • Hypertension	Lack of exercise, unhealthy eating habits, alcohol consumption, smoking, use of drug, excess weight, obesity, and mental stress.
Occupational diseases	Silicosis	Breathing in silica dust (miners and sand workers).
Autoimmune disorder	Rheumatoid arthritis	In some people, certain immune cells damage the cartilage and the synovial lining.
Nutritional disorders	Anaemia	Iron deficiency.

Hereditary diseases

- Diseases caused by genetic defects are called hereditary diseases.

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Diseases	Reason	Mode of Infection
Sickle cell anaemia	Defect of the gene that controls the production of haemoglobin	• Red blood cells become sickle-shaped. • The deformed red blood cells get stuck in capillaries and block blood flow.
Haemophilia	Defect of the gene that controls the production of proteins for blood clotting	• Absence of the protein that helps in blood clotting. • Even small wounds cause excessive bleeding.

Treatment

- Ayurveda – Charaka
- Homeopathy – Samuel Hahnemann
- Modern medicine – Hippocrates

Methods followed by modern medicine

Diagnosis

- Assessing the symptoms
- Laboratory test
- Body check-up

Treatment

- Medicine
- Surgery
- Organ transplantation
- Radiation therapy

Diagnostic equipment

Stethescope	To identify sound variations inside the body (heart, lungs, intestines).
Sphygmomanometer	To measure blood pressure.
Digital B.P apparatus	To measure blood pressure.
Pulse oximeter	To measure blood oxygen saturation level and heart rate.
Thermometer	To measure body temperature.
Endoscope	Used for visual examination of the internal parts of body cavities and organs.
US scanner, CT scanner, MRI scanner	To record the defect of internal organs.

Specialization.	**Related field**
General medicine	Prevention, diagnosis, and treatment of various diseases.
Cardiology	Treatment of heart and blood vessels.

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Paediatrics	Treatment of diseases in children.
Gynaecology	Treatment for female reproductive system.
Nephrology	Treatment of kidney diseases.
Neurology	Treatment of nerve diseases
Urology	Treatment of urinary tract diseases
ENT	Treatment of ear, nose, and throat diseases.

Medicine

- Medicines may be derived from plants, animals or microorganisms or they may be artificially synthesised.

  Eg : Antibiotics

Antibiotics

- Antibiotics are medicines that are either derived from microorganisms or produced synthetically, and are used against bacteria.

Side effects of antibiotics

1. Problems in the digestive system such as diarrhoea, vomiting and stomach pain.
2. Allergy
3. Damage to the liver and kidney.
4. Destruction of beneficial bacteria in the intestine.
5. Resistance against antibiotics (Antibiotic resistance)

Different types of medicines

Bacteria - Antibiotic
Virus – Antiviral
Fungus - Antifungal
Protozoa - Antiprotozoal
Worms - Antihelminthic
Fever - Antipyretic

Guidelines for blood donors

1. Individuals between the age of 18 and 60 can donate blood.
2. A minimum weight of 50kg is required.
3. People with conditions such as asthma, epilepsy ,heart disease or cancer , are not allowed to donate blood.

Blood transfusion

- The process of transferring blood or its components from one person to another is called blood transfusion.
- Blood groups are determined by the presence of A antigen and B antigen present on the surface of RBCs.
- If D antigen is present, the blood group is called positive; if absent, it is called negative.
- If the donor’s blood contains an antigen that is naturally absent in the recipient’s blood, the antibodies present in the recipient’s plasma may act against it, leading to agglutination of the blood.

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Basis of blood groups

Blood group	Antigen	Antibody
A	A	b
B	B	a
AB	A , B	nil
O	nil	a , b

Animal Diseases

Disease	Pathogen	Affects	Prevention
Foot and mouth disease	Virus	Cattle (Cows, buffaloes.., etc.)	Vaccine
Mastitis	Bacteria	Cattle	Vaccine
Anthrax	Bacteria	Cattle	Vaccine
Ranikhet disease	Virus	Chicken, duck	Vaccine

Plant disease

Disease	Pathogen	Symptoms
Wilt of Brinjal	Bacteria	The young leaves wilt quickly. Gradually, the leaves turn yellow, growth becomes stunted and the branches are reduced in number.
Blight disease of paddy	Bacteria	Round yellow spots on the leaves, spots on the rice grain.
Bunchy top of Banana	Virus	Leaves become short, narrow and stand upright, growth becomes stunted.
Mosaic disease in Tapioca	Virus	Initially light yellow coloured patches are formed on the leaves. Gradually these become white. Growth retards and the size of tapioca decreases.
Bud rot of coconut	Fungus	The tender leaves turn into yellow, and the crown of the coconut tree droops and falls.
Quick wilt of pepper	Fungus	Wet marks on the leaves and stem. Leaves fall and the plant wilts quickly.

Immunity mechanisms in plants

- A rigid cell wall, epidermis and waxy coating on leaves and stems prevent the entry of pathogens.
- When one part of the plant is attacked, it sends a warning signals from cell to cell about the diseases .
- If pathogen enters, a polysaccharide called **callose** prevents their spread.
- Enzymes break down the cell walls of pathogens.
- The cells in the infected area undergo autolysis to prevent the spread of the disease.
- Antimicrobial substances either kill pathogens or inhibit their growth.

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Unit-6

Biology and Technology

Biotechnology and Genetic Engineering

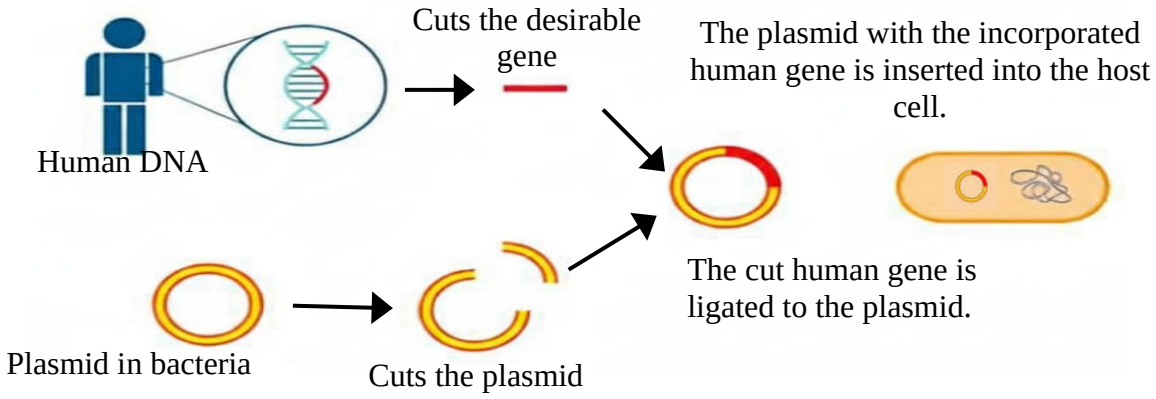
- Biotechnology is the technology of using living things or their parts, to provide products and services that are useful to humans.
- Genetic engineering is the technology for creating desired traits in organisms by combining or deleting genes.

Techniques used in genetic engineering

1. Recombinant DNA technology

- Recombinant DNA technology combines DNA segments from two or more different organisms to form recombinant DNA.

Genetic scissors : Restriction Endonuclease (Enzyme used to cut specific gene).
Genetic glue : Ligase (Enzyme that joins two DNA segments)
Vector : Plasmid in bacteria (Circular DNA) , some viruses carry genes.



1.The required gene is cut out from human DNA.
2.The bacterial plasmid, DNA is cut.
3.The cut human gene is ligated to the plasmid.
4.The plasmid with the incorporated human gene is inserted into the host cell.
5.As the inserted genes become active in the cell,the desired products are formed.

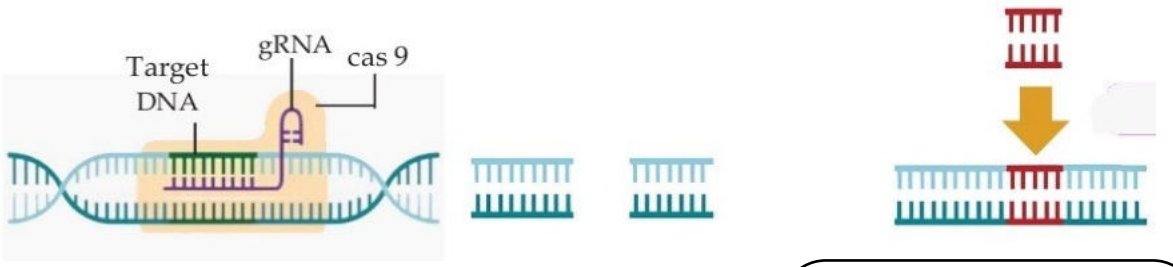
CRISPR technology

- CRISPR is a technology used to edit DNA.
- The CRIPR technology was designed from a natural defense mechanism that bacteria use, to defend against viruses.
- It uses an enzyme called Cas 9 as scissors to cut DNA.

Guide RNA : An RNA that identifies the DNA segment to be cut.

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CRISPR TECHNOLOGY



Identifying the target: The guide RNA precisely identifies the DNA segment that needs to be cut.

Cutting: The Cas9 enzyme cuts the DNA segment identified by the guide RNA

Editing:

DNA is edited by making the required changes.
This helps to remove unwanted traits or add new traits.

Groups	Characteristics	Examples
Plants	Pest resistance	Bt. Cotton
	Weedicide resistance	Soya bean
	Virus resistance	RSV resistant papaya
	Nutrient resistance	Golden rice
Animals	Fast growth, Diseases resistance, High productivity	Salmon fish
	Fluorescence	Glo fish
Micro-organisms	Ability to decompose waste.	Pseudomonas putida
	Ability to produce insulin	E-coli
	Fermentation	Baker's yeast
	High nitrogen fixing ability	Rhizobium

Human genome project - HGP

- Genome is the sum total of all the genetic materials in an organism.
- The Human Genome Project was a global scientific initiative that began in 1990 and was completed in 2003, with the goal of completely understanding the information in the human genome.
- Gene mapping technology which identifies the location of genes in DNA.

Major findings of the Human Genome Project

- The exact sequence of 300 crores base pairs in the human genome.
- Humans have about 20,000 to 25,000 genes.

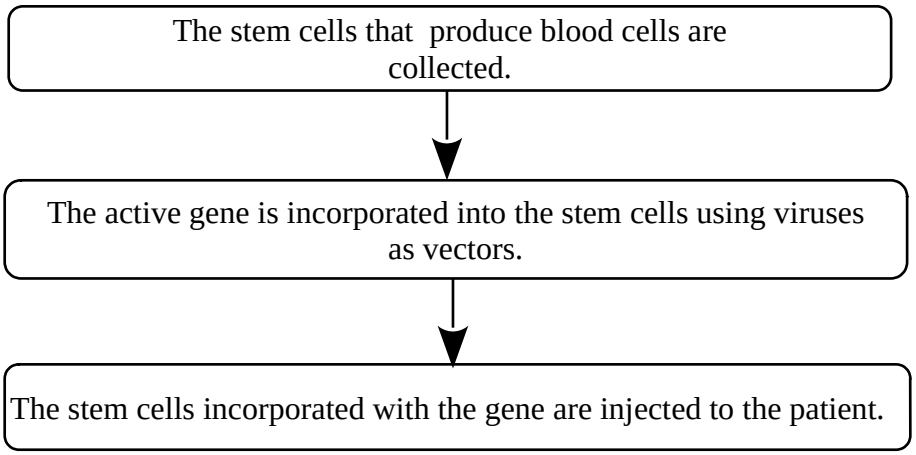
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- Only about 1–2% of the human genome consists of functional genes.
- The remaining bulk part is known as junk DNA .

Gene therapy

- Gene therapy is the method of treating diseases by removing the defective genes and replacing them with functional genes.
- For example, stem cells can be used to completely cure Sickle cell anaemia.

Stages of stem cell therapy



DNA Fingerprinting

- DNA finger printing was discovered by scientist Alec Jeffrey
- The technology that is used to analyse the sequence of nucleotides in DNA and to identify individuals accordingly is called **DNA fingerprinting**.
- 99.9% of DNA is the same in all humans, variations are seen in the remaining 0.1% .
- Fingerprints that vary from person to person, the sequence of nucleotides in each person also vary. This peculiarity helps to identify individuals.
- Parents, offsprings, and siblings share more resemblances in DNA pattern.

Uses of DNA Fingerprinting

- To identify culprits
- To identify genetic disorders and learn about hereditary diseases
- To determine the biological relationship between parents and offspring
- To understand lineage by identifying fossils
- To identify victims of accidents or disasters
- To track endangered species.

Human Microbiome project(HMP)

- The human microbiome is the collection of microorganisms and their genes that inhabit and interact within the human body.
- It is a community of microorganisms that includes bacteria, archaea, fungi and viruses.
- They provide us with many services, including immunity,digestion, and the production and absorption of vitamins.

Services provided by the Human Microbiome Project

- HMP proved that the human body is composed of crores of microorganisms and their co-existence is essential for our well being.

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- Cause of disease is not only the pathogens but the change in the equilibrium of microorganisms in the body can lead to illness.
- HMP has opened new ways for disease diagnosis and treatment.
- This project opened a great opportunity to design personalised medicine and to arrange treatment according to each individual's microbiome.

Biology and Artificial intelligence

Services provided by artificial intelligence in the field of biology

- AI plays a major role in many areas of biological study, including disease diagnosis, drug discovery and genetic engineering.
- It helps in discovering new drugs, for predicting the effectiveness of chemical substances and recommendation of the most suitable treatment.
- Personalised medicine analyses each individual’s genetic makeup and health information and recommends the most suitable treatment regimens for them.

Genome sequencing

- Utilised in gene editing technology for the treatment of genetic disorders.

Technologies suitable for environmental protection

- **Wildlife Tracking Sensors**

  These help to plan conservation strategies by monitoring the movement, behaviour, and location of wild animals.
- **Global Biodiversity Data**

  A technology that collects information about biodiversity using satellite remote sensing, acoustic analysis and camera traps.

- **Cryopreservation**

  This is the process of preserving living cells, tissues and organs at very low temperatures

  (-196°C).

- **Bioremediation**

  Uses microorganisms and plants to remove pollutants from contaminated environments such as soil, water, and air. This converts the pollutants into harmless compounds.